

Report Guidelines

Indian Institute of Technology Delhi

Course: CLL231

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Self-assessment*

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Problem Statement

The objective of this study is to analyze turbulent flow behavior in a straight square conduit. The conduit has a side length of 0.05 mm and a total length of 0.5 mm. Water is considered as the working fluid, and the flow is assumed to be steady, three-dimensional, and turbulent in nature.

The normalized velocity profile $\left(\frac{u}{u_{\max}}\right)$ is plotted as a function of $\frac{y}{H}$ and compared with the theoretical velocity distribution given by best fitted with cubic equations which are written on the graphs.

Since the Reynolds number exceeds 4000 in all cases, the flow is confirmed to be turbulent.

The variation of pressure drop with maximum velocity is also studied for five different Reynolds numbers:

$$\{4000, 4250, 4500, 4750, 5000\}$$

According to the Reynolds number above the corresponding velocities (u_1, u_2, u_3, u_4, u_5) are respectively:

$$\{80, 85, 90, 95, 100\} \text{ m/s}$$

Relation Between Velocity and Reynolds Number

The Reynolds number is defined as:

$$\text{Re} = \frac{\rho u D_h}{\mu}$$

Rearranging:

$$u = \frac{\text{Re} \mu}{\rho D_h}$$

Substituting:

$$\rho = 1000, \quad \mu = 0.001, \quad D_h = 5 \times 10^{-5} \text{ m}$$

$$u = \frac{\text{Re}}{50}$$

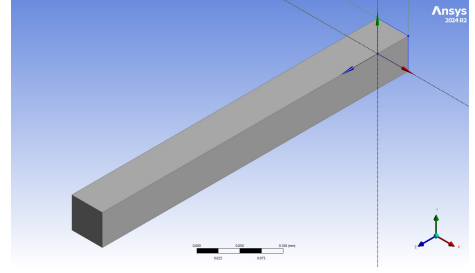


Figure 1: Square conduit geometry

Numerical Simulation Methodology

A CFD approach was used to simulate turbulent flow. The properties used are: $\rho = 1000 \text{ kg/m}^3$, $\mu = 0.001 \text{ Pa}\cdot\text{s}$, and $D_h = 0.05 \text{ mm}$.

A base velocity of 80 m/s was used for mesh independence testing.

Three meshes were used:

- M1 (Fine): 50 divisions
- M2 (Medium): 20 divisions
- M3 (Coarse): 10 divisions

Cell sizes:

- M1: $1 \mu\text{m}$
- M2: $2.5 \mu\text{m}$
- M3: $5 \mu\text{m}$

A graph of ΔP vs number of mesh elements was plotted to ensure mesh independence.

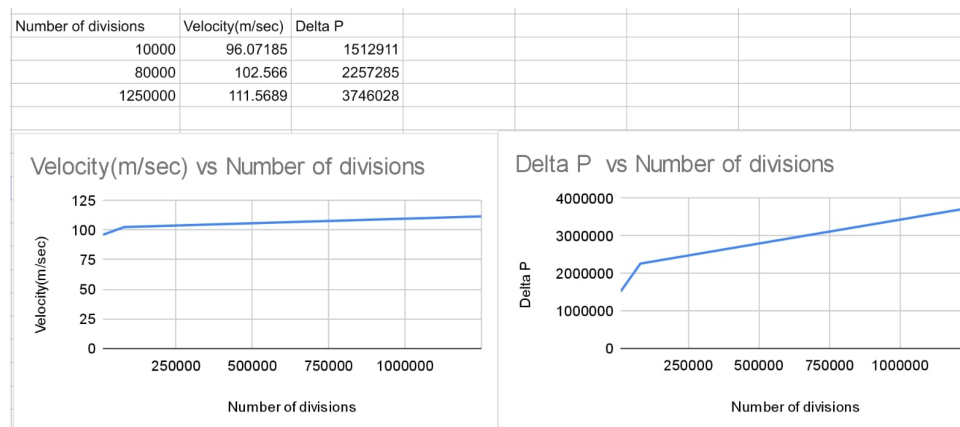


Figure 2: Mesh independence test: ΔP vs Number of mesh elements

Velocity Profile Analysis

Velocity profiles $\left(\frac{u}{u_{\max}}\right)$ vs $\frac{y}{H}$

- Graphs for comparison of theoretical and simulated profile are below:

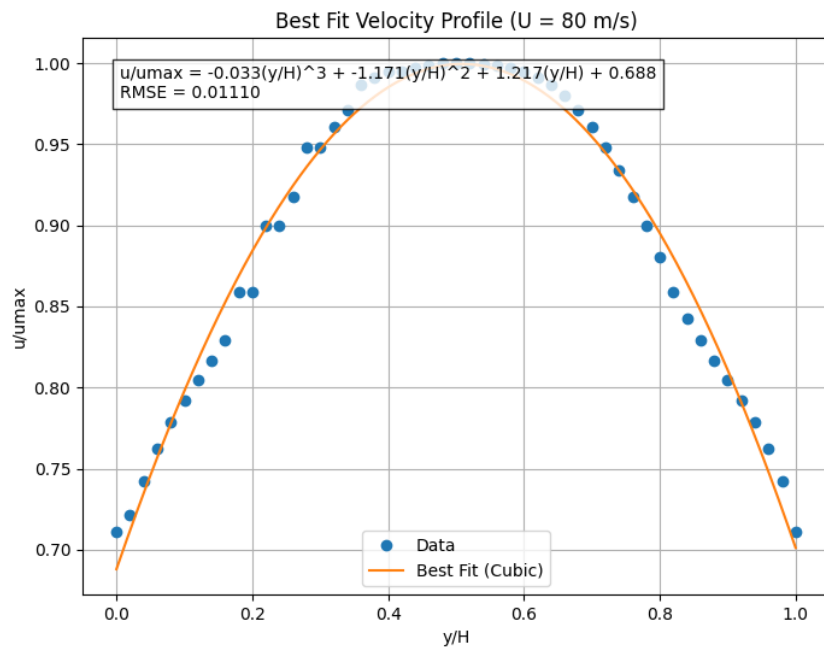


Figure 3: Normalized velocity profile vs y/H for $U=80$ m/sec

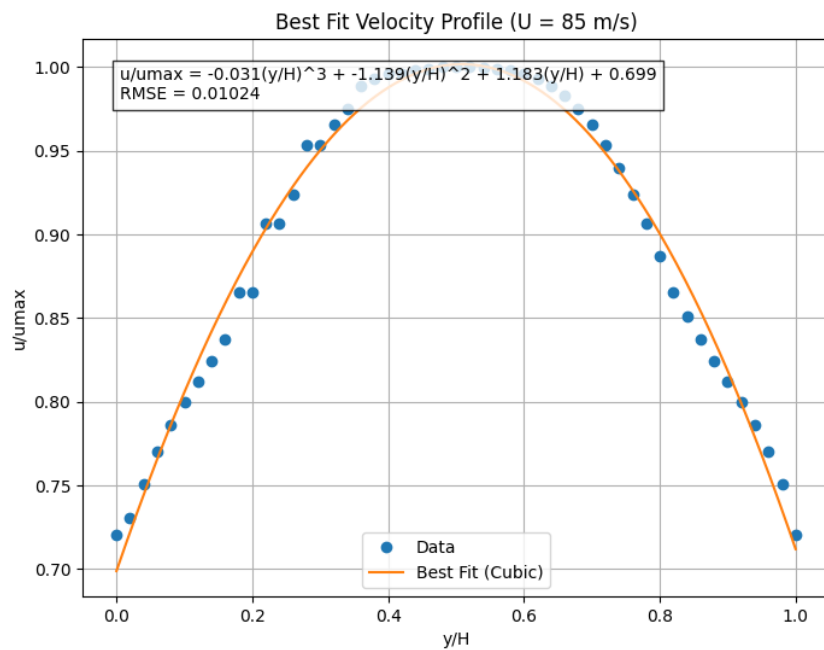


Figure 4: Normalized velocity profile vs y/H for $u=85$ m/sec

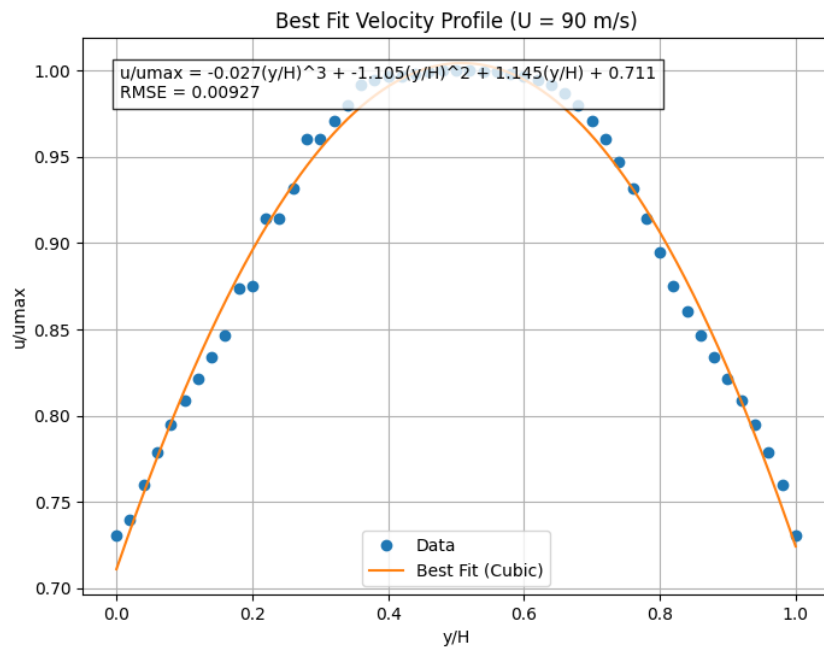


Figure 5: Normalized velocity profile vs y/H for 90 m/sec

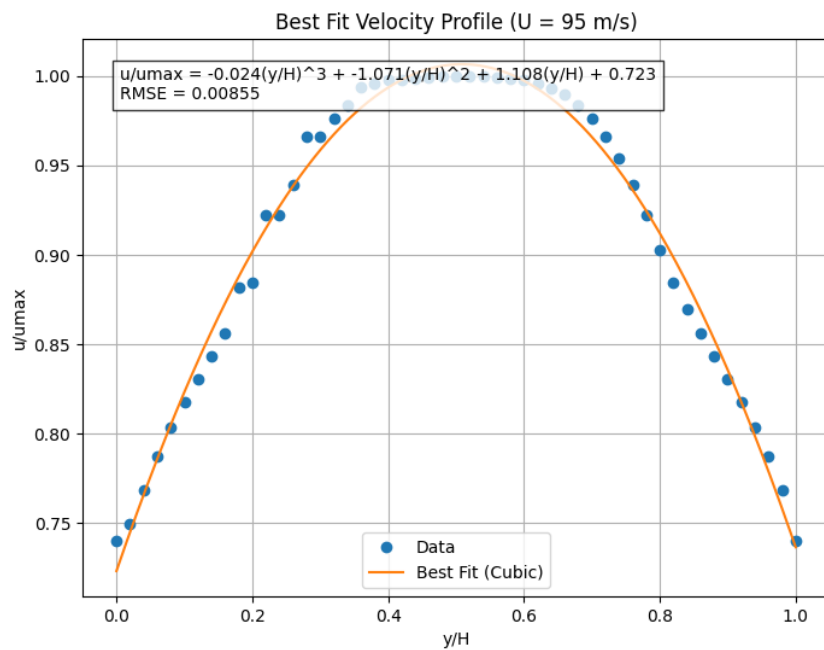


Figure 6: Normalized velocity profile vs y/H for 95 m/sec

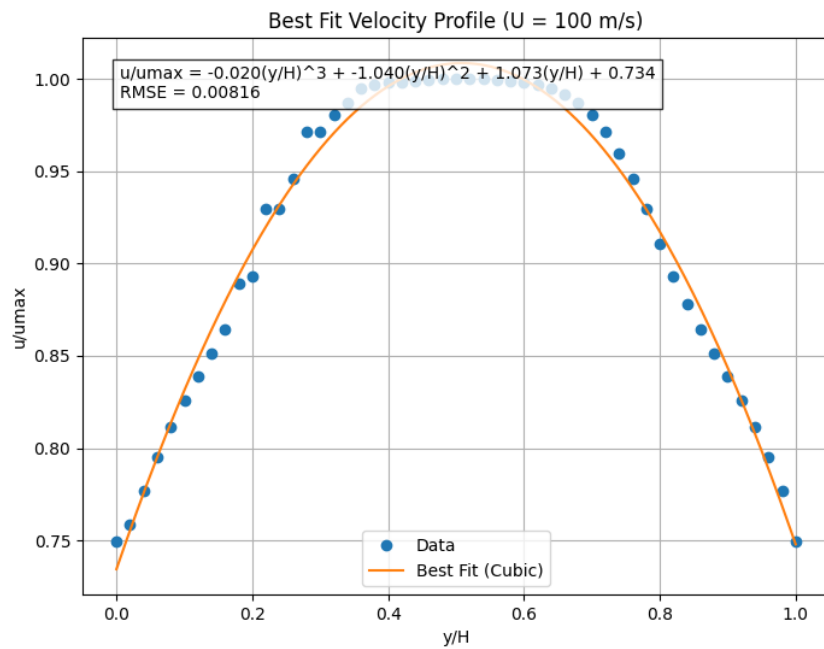


Figure 7: Normalized velocity profile vs y/H 100 m/s

Pressure Drop Analysis

The variation of pressure drop with maximum velocity was plotted.

Additionally, ΔP vs mesh elements was plotted during mesh independence testing.

These results confirm that pressure drop increases with velocity and validate turbulent flow behavior.

u	umax(m/s)	delta P (Pa)
80	111.5689	3746028
85	117.5835	4059470
90	123.536	4377920
95	129.4556	4701408
100	135.3074	5029087

Figure 8: Data for the below graph

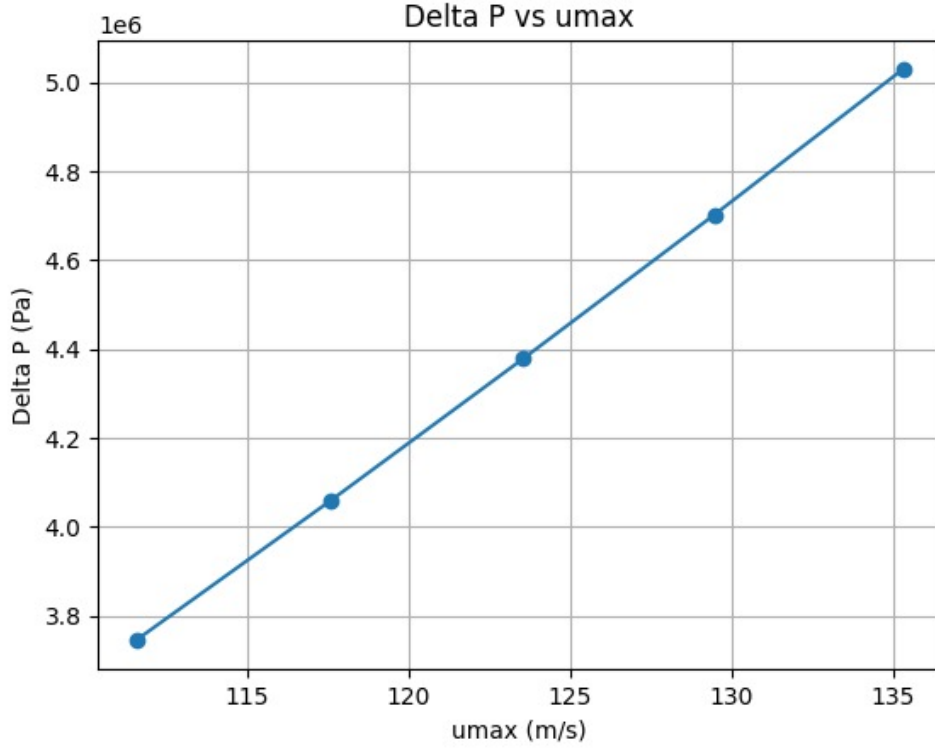


Figure 9: Variation of pressure drop (ΔP) with maximum velocity(U_{max})

Results and Discussion

3.1 Mesh Independence Study

Before analyzing the flow physics, a mesh independence study was conducted at a base inlet velocity of 80 m/s using three progressively refined meshes: M3 (Coarse, 10 divisions, ~10,000 elements), M2 (Medium, 20 divisions, ~80,000 elements), and M1 (Fine, 50 divisions, ~1,250,000 elements). The pressure drop and maximum centerline velocity were monitored across all three meshes, and the results are summarized below:

Mesh	Divisions	Max Velocity (m/s)	ΔP (Pa)
M3 (Coarse)	10,000	96.072	1,512,911
M2 (Medium)	80,000	102.566	2,257,285
M1 (Fine)	1,250,000	111.569	3,746,028

As seen in Figure 2, both the maximum velocity and the pressure drop increase significantly from the coarse to the medium mesh, but the rate of change diminishes as the mesh is further refined toward M1. This asymptotic behavior indicates that the numerical solution is approaching convergence and is no longer sensitive to mesh refinement beyond 1,250,000 elements. Consequently, the fine mesh M1 was selected for all subsequent simulations as it provides the best balance between accuracy and computational cost.

3.2 Normalized Velocity Profile and Cubic Curve Fitting

The normalized velocity profiles (u/u_{max}) were obtained from the CFD simulation and plotted against the normalized transverse coordinate (y/H) for all five inlet velocities. To characterize the shape of the

simulated profile quantitatively, a cubic polynomial of the form

$$\frac{u}{u_{\max}} = a \left(\frac{y}{H} \right)^3 + b \left(\frac{y}{H} \right)^2 + c \left(\frac{y}{H} \right) + d$$

was fitted to the simulation data using a least-squares approach. The fitted coefficients and the corresponding Root Mean Square Error (RMSE) for each case are listed in Table 2.

u (m/s)	a	b	c	d	RMSE
80	−0.033	−1.171	1.217	0.688	0.01110
85	−0.031	−1.139	1.183	0.699	0.01024
90	−0.027	−1.105	1.145	0.711	0.00927
95	−0.024	−1.071	1.108	0.723	0.00855
100	−0.020	−1.040	1.073	0.734	0.00816

*tableTable 2: Cubic fit coefficients and

RMSE for each inlet velocity

The cubic fits (shown as the orange curves in Figures 3–7) track the simulated data (blue dots) very closely across all five cases, with RMSE values ranging from 0.01110 at $u = 80$ m/s down to 0.00816 at $u = 100$ m/s. The consistently low RMSE confirms that the cubic polynomial is an excellent descriptor of the simulated turbulent velocity profile shape across this Reynolds number range.

Physically, the profiles exhibit the characteristic flat-topped shape of turbulent flow: velocity rises steeply near both walls ($y/H \approx 0$ and $y/H \approx 1$) due to the no-slip boundary condition, and forms a broad, nearly uniform plateau in the central core of the conduit. The peak normalized velocity of $u/u_{\max} = 1.00$ is attained at approximately $y/H \approx 0.5$, confirming the expected symmetry of the flow about the conduit centerline.

A notable trend observed in the cubic coefficients is that the magnitude of coefficient a (the cubic term) decreases with increasing inlet velocity, from −0.033 at 80 m/s to −0.020 at 100 m/s. Similarly, coefficient b becomes less negative and d increases slightly. This systematic variation indicates that as the Reynolds number increases from 4000 to 5000, the velocity profile becomes progressively flatter in the core region and the near-wall gradients become somewhat steeper — consistent with the well-known physical behavior that higher Reynolds number turbulent flows develop a fuller velocity profile due to more vigorous turbulent mixing. The RMSE also decreases monotonically with increasing Reynolds number, suggesting that the cubic model fits the profile shape better at higher velocities, likely because the profile becomes smoother and more symmetric.

3.3 Pressure Drop Variation with Maximum Velocity

The pressure drop (ΔP) across the conduit was recorded for all five simulated conditions and is tabulated below along with the corresponding maximum centerline velocities:

Inlet Velocity u (m/s)	Reynolds Number	$u_{\max}$ (m/s)	ΔP (Pa)
80	4000	111.569	3,746,028
85	4250	117.584	4,059,470
90	4500	123.536	4,377,920
95	4750	129.456	4,701,408
100	5000	135.307	5,029,087

*tableTable 3: Pressure drop

and maximum velocity for each Reynolds number

The results clearly show that ΔP increases monotonically with increasing inlet velocity. As the inlet velocity rises from 80 m/s to 100 m/s, the pressure drop increases from approximately 3.75 MPa to 5.03 MPa — a total rise of roughly 34% over the studied velocity range.

As seen in Figure 9, the ΔP vs $u_{\max}$ relationship appears nearly linear over this range. This is consistent with turbulent flow theory: for turbulent flow in a conduit, the Darcy–Weisbach equation gives

$$\Delta P = f \cdot \frac{L}{D_h} \cdot \frac{\rho u^2}{2}$$

where the friction factor f itself varies with Reynolds number according to correlations such as the Blasius formula ($f \propto \text{Re}^{-0.25}$). Over the narrow Reynolds number range of 4000–5000 studied here, f changes only marginally, making the ΔP – $u_{\max}$ relationship appear approximately linear rather than strictly quadratic. The maximum centerline velocity $u_{\max}$ also scales proportionally with the inlet velocity u , with the ratio $u_{\max}/u \approx 1.35$ remaining nearly constant across all cases — a value physically consistent with turbulent flow in a square conduit where the centerline velocity exceeds the bulk mean velocity.

3.4 Overall Interpretation

The CFD results collectively demonstrate that the simulation successfully captures the key features of turbulent flow in a square conduit across the Reynolds number range of 4000 to 5000. The mesh independence study establishes M1 as the appropriate mesh for accurate and converged predictions. The cubic polynomial fits provide a compact and accurate mathematical description of the velocity profile at each Reynolds number, with RMSE values below 0.012 in all cases. The systematic variation of the cubic coefficients with Reynolds number reflects the physical flattening of the turbulent velocity profile at higher flow rates. Finally, the pressure drop data confirm the expected monotonic increase with velocity and are consistent with the Darcy–Weisbach framework for turbulent conduit flow. Taken together, these results validate the numerical methodology and confirm that the simulation faithfully reproduces turbulent flow physics.